

SAT Math Mastery Program · Week 1, Session A

Linear Equations: Structure, Rearranging & Translating

Time	120 minutes: 90 min solving + 30 min self-check and error log
Target topics	one-variable equations · solution-count structure · identities & coefficient matching · literal equations · word translation · inequalities
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Name / Date	_____

1 · SKILLS & IDEAS

- S1.** Solve any linear equation (distribute, LCD for fractions/decimals, collect, divide).
- S2.** Spot a repeated *block* and solve for the block instead of the variable.
- S3.** Decide *one* / *none* / *infinitely many* solutions by comparing coefficients, before solving.
- S4.** Match coefficients when an equation is true for every x ; solve for constants.
- S5.** Rearrange literal equations (make any letter the subject).
- S6.** Translate words: “less than”, consecutive integers, ratios, fixed-plus-rate.
- S7.** Linear inequalities: sign flip, compound and *or*-unions, integer counts, rounding to a feasible whole number.
- S8.** Parameter equations: the solution depends on a constant; integer-solution and case-analysis conditions.

2 · REQUIRED FACTS

- Equal operations on both sides keep an equation true; **multiplying or dividing an inequality by a negative reverses it.**
- “ n less than m ” = $m - n$; “at least” $\geq$; “at most” / “no more than” $\leq$; “more than” $>$.
- Consecutive integers differ by 1; consecutive *even or odd* integers differ by 2.
- A ratio $a : b = m : n$ means $a = mt$, $b = nt$; the whole is $(m + n)t$.
- Integers with $p \leq x \leq q$: $q - p + 1$ of them; drop one for each open end.
- Isosceles: two sides equal; every side must be > 0 and satisfy $a + b > c$.
- $\frac{p}{q}$ is an integer $\iff q \mid p$ (for integers p, q , $q \neq 0$).

3 · REQUIRED FORMULAS

$$ax + b = cx + d \implies (a - c)x = d - b, \quad x = \frac{d - b}{a - c} \quad (a \neq c)$$

$$P_{\text{rect}} = 2(\ell + w), \quad A_{\text{trap}} = \frac{1}{2}h(b_1 + b_2), \quad F = \frac{9}{5}C + 32$$

$$n, n + 2, n + 4, n + 6 \text{ (consecutive odd),} \quad \text{sum of } k \text{ terms} = k \cdot (\text{mean})$$

4 · MINI THEORY SHEET

Master reduction. Every linear equation collapses to $mx = n$. Then:

$m \neq 0$	one solution	$x = \frac{n}{m}$
$m = 0, n \neq 0$	no solution	$0 = n$ is false
$m = 0, n = 0$	infinitely many	$0 = 0$ is true for all x

Two-parameter identity. If $p(x+q) = rx + s$ holds for all x : coefficients of x match ($p = r$) and constants match ($pq = s$).

Block rule. If an expression E occurs, $cE = k \Rightarrow E = \frac{k}{c}$; scale or shift the block, never expand it.

Inequalities. Same steps as equations; reverse only on $\times/\div$ by a negative. Compound: operate on all three parts. And = overlap, or = union. Closed dot $\leq, \geq$; open dot $<, >$.

Contexts. Fixed + rate: $y = \text{fee} + \text{rate} \cdot n$. For “at most” counts solve, then *round down*; for “at least” counts round up.

Case analysis. If a condition (isosceles, equal sums) can occur in several pairings, solve each pairing, then reject any that break positivity or the triangle inequality.

Parameter solution. Solve for x in terms of the constant first; then impose the condition on that expression.

5 · Strategy Box

Shortcuts. (1) *Block*: if the target is a multiple of an expression you were given, scale—don’t solve. (2) *Structure first*: “how many solutions”, “no solution”, “infinitely many” $\Rightarrow$ compare coefficients, zero algebra. (3) *Back-solve*: plug options into the original equation, middle values first.

Traps. Sign of a distributed negative · forgetting to reverse an inequality · “less than” written backwards · and vs. or · open vs. closed endpoints · answering for x when the question asks for k , $x - 3$, area, or a product · rounding the wrong way in counting contexts.

Calculator (Desmos). Enter each side as y_1, y_2 : intersections = solutions; parallel = none; identical = infinite. Use a slider for a parameter. Keep fractions exact; never round mid-solution.

Timing. Easy ≤ 1.5 min · Medium ≤ 3 · Hard ≤ 4.5 · Challenge ≤ 7 (total ≈ 80 min, 10 min buffer). If stuck $>$ the limit, mark and return.

Pattern cues. “for all x ” $\rightarrow$ match coefficients · “true for infinitely many” $\rightarrow$ same slope and same constant · “what is (expr)?” $\rightarrow$ block · “greatest/least whole number” $\rightarrow$ inequality then round.

6 · GUIDED EXAMPLES

Example 1 — Structure first (parameter)

For what value of k does $4(x - 2) + k = 4x - 3$ have (a) no solution, (b) infinitely many solutions?

Expand: $4x - 8 + k = 4x - 3$. The x -coefficients agree ($4 = 4$), so only the constants decide: compare $k - 8$ with -3 .

(b) infinitely many iff $k - 8 = -3$, so $k = 5$. (a) none iff $k - 8 \neq -3$, i.e. **every** $k \neq 5$.

Check: $k = 5$ gives $4x - 3 = 4x - 3$ (true always). $k = 0$ gives $4x - 8 = 4x - 3$ (false always).

Example 2 — Fractions: clear with the LCD

Solve $\frac{x+1}{2} - \frac{x-3}{5} = 2$.

LCD = 10: $5(x+1) - 2(x-3) = 20 \Rightarrow 5x + 5 - 2x + 6 = 20 \Rightarrow 3x + 11 = 20 \Rightarrow x = 3$.

Check: $\frac{4}{2} - \frac{0}{5} = 2$. $\checkmark$ (Note the minus sign distributes to *both* terms of $x - 3$.)

Example 3 — Words $\rightarrow$ equation (consecutive integers)

The sum of three consecutive even integers is 54. What is the largest?

Let them be $n, n + 2, n + 4$: $3n + 6 = 54 \Rightarrow n = 16$. Integers: 16, 18, 20. Largest = **20**.

Shortcut: $54 \div 3 = 18$ is the middle term; add 2.

7 · PRACTICE SET — 24 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Write your work in the space below each problem. Circle your final answer. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

EASY · NO CALCULATOR NEEDED

If $4x + 3 = 27$, what is the value of $8x + 6$?

(A) 33

(B) 48

(C) 54

(D) 60

2.

EASY · NO CALCULATOR NEEDED

What is the solution to $-2(3x - 5) = 4x - 10$?

(A) -2

(B) 0

(C) 2

(D) 20

3.

EASY · NO CALCULATOR NEEDED

What is the solution to $\frac{2x+1}{3} = \frac{x-4}{2}$?

(A) -14

(B) -2

(C) 2

(D) 14

4.

EASY · NO CALCULATOR NEEDED

The formula $F = \frac{9}{5}C + 32$ converts a temperature from degrees Celsius C to degrees Fahrenheit F . Which equation gives C in terms of F ?

(A) $C = \frac{5}{9}(F - 32)$ (B) $C = \frac{5}{9}F - 32$ (C) $C = \frac{9}{5}(F - 32)$ (D) $C = \frac{5}{9}F + 32$

5.

EASY · NO CALCULATOR NEEDED

Three less than twice a number is five more than the number. What is the number?

6.

EASY · NO CALCULATOR NEEDED

What is the least integer x that satisfies $-3x + 7 \leq 22$?

7.

MEDIUM · NO CALCULATOR NEEDED

For what value of k does the equation $\frac{1}{2}(6x + k) = 3x + 7$ have infinitely many solutions?

8.

MEDIUM · DESMOS

In the equation $(a - 2)x + 3 = 5x + 7$, a is a constant. If the equation has no solution, what is the value of a ?

(A) 2

(B) 3

(C) 5

(D) 7

9. MEDIUM · NO CALCULATOR NEEDED

If $3(2x + a) = bx + 21$ is true for all values of x , where a and b are constants, what is $a + b$?

10. MEDIUM · NO CALCULATOR NEEDED

The Dar Alfikr robotics club has girls and boys in the ratio 3 : 5. There are 96 members. How many more boys than girls are in the club?

- (A) 12 (B) 24
(C) 36 (D) 60

11. MEDIUM · NO CALCULATOR NEEDED

The area of a trapezoid is $A = \frac{1}{2}h(b_1 + b_2)$. Which expression equals b_1 ?

- (A) $\frac{2A - b_2}{2h}$ (B) $\frac{A}{2h} - b_2$
(C) $\frac{2A}{h + b_2}$ (D) $\frac{2A}{h} - b_2$

12.

MEDIUM · CALCULATOR

A tutoring academy charges a one-time registration fee of 150 SAR plus 45 SAR per session. Ahmad paid a total of 690 SAR. How many sessions did he attend?

- (A) 12 (B) 15
(C) 19 (D) 540

13.

MEDIUM · CALCULATOR

A Riyadh Metro card holds 60 SAR. Each trip costs 4.5 SAR. Sara wants at least 5 SAR to remain on the card. What is the greatest number of trips she can take?

- (A) 11 (B) 12
(C) 13 (D) 14

14.

MEDIUM · NO CALCULATOR NEEDED

How many integers x satisfy $-7 < 3 - 2x \leq 9$?

- (A) 7 (B) 8
(C) 9 (D) 10

15.

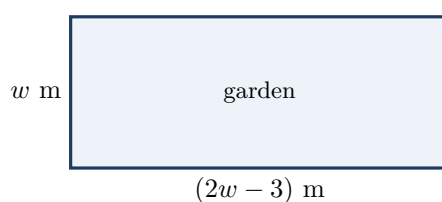
MEDIUM · CALCULATOR

What is the solution to $0.4(x + 10) = 0.25x + 7.6$?

16.

MEDIUM · NO CALCULATOR NEEDED

The figure shows a rectangular school garden with width w meters and length $(2w - 3)$ meters. Its perimeter is 54 meters. What is its area, in square meters?



17.

HARD · DESMOS

The equation $(a^2 - 9)x = a + 3$ has a real constant a . For how many real values of a does the equation *not* have exactly one solution?

- (A) 0 (B) 1
(C) 2 (D) 3

18.

HARD · NO CALCULATOR NEEDED

If $\frac{a+b}{a-b} = 3$ and $b \neq 0$, what is the value of $\frac{a}{b}$?

- (A) 1 (B) $\frac{3}{2}$
(C) 2 (D) 3

19.

HARD · NO CALCULATOR NEEDED

The sum of four consecutive odd integers is 18 more than twice the largest of the four. What is the product of the two middle integers?

20.

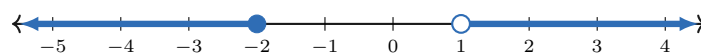
HARD · CALCULATOR

What is the solution to $\frac{2x-1}{3} - \frac{x+2}{4} = \frac{x}{6} - \frac{1}{2}$?

21.

HARD · NO CALCULATOR NEEDED

The graph shows the solution set of a compound inequality on the number line. Which of the following has this solution set?



(A) $3x + 1 \geq -5$ **or** $x - 4 < -3$

(B) $3x + 1 \leq -5$ **and** $x - 4 > -3$

(C) $3x + 1 < -5$ **or** $x - 4 \geq -3$

(D) $3x + 1 \leq -5$ **or** $x - 4 > -3$

22.

HARD · NO CALCULATOR NEEDED

The solution to $\frac{x + k}{4} = 3$ is equal to $2k$. What is the value of the constant k ?

SAT CHALLENGE • CALCULATOR

A triangle with side lengths labeled as $x + 2$, $5x - 16$, and $3x - 2$.

8 · HINT SECTION — HINTS ONLY, NO SOLUTIONS

1. $8x + 6$ is a multiple of $4x + 3$.
2. Distribute the -2 to *both* terms.
3. Cross-multiply.
4. Undo $+32$ first, then undo $\times \frac{9}{5}$.
5. “Three less than $2n$ ” is $2n - 3$.
6. Dividing by -3 reverses the inequality.
7. After distributing, the x -terms already match; make the constants match.
8. No solution: equal x -coefficients, different constants.
9. Expand the left side, then match x -coefficients and constants.
10. Total parts = 8; find one part.
11. Clear the $\frac{1}{2}$ and the h first.
12. Subtract the fee *before* dividing by the rate.
13. $60 - 4.5t \geq 5$; the count of trips must be a whole number.
14. Subtract 3 from all three parts, then divide by -2 and flip *both* signs.
15. Distribute, then collect the x -terms on the left.
16. Write the perimeter equation, find w , then the area.
17. Exactly one solution needs the x -coefficient $\neq 0$. Test each value that makes it 0.
18. Clear the denominator, then divide by b .
19. Let the integers be $n, n + 2, n + 4, n + 6$.
20. LCD is 12; multiply every term, including $\frac{1}{2}$.
21. Find each piece’s solution set; *or* means union.
22. Solve for x in terms of k , then set it equal to $2k$.
23. Solve for x as a fraction in k , split off a whole number, then use divisibility. Check $k = 1$ separately.
24. Three pairings. Solve each, then reject any with a non-positive side or a violated triangle inequality.

9 · FULL SOLUTIONS

1. **Answer: (C).** $8x + 6 = 2(4x + 3) = 2(27) = 54$. (Solving: $x = 6$, $8(6) + 6 = 54$.)
2. **Answer: (C).** $-6x + 10 = 4x - 10 \Rightarrow 20 = 10x \Rightarrow x = 2$.
3. **Answer: (A).** $2(2x + 1) = 3(x - 4) \Rightarrow 4x + 2 = 3x - 12 \Rightarrow x = -14$.
4. **Answer: (A).** $F - 32 = \frac{9}{5}C \Rightarrow C = \frac{5}{9}(F - 32)$.
5. **Answer: 8.** $2n - 3 = n + 5 \Rightarrow n = 8$.
6. **Answer: -5.** $-3x \leq 15 \Rightarrow x \geq -5$. Least integer: -5 .
7. **Answer: 14.** $3x + \frac{k}{2} = 3x + 7$. Infinitely many iff $\frac{k}{2} = 7$, so $k = 14$.
8. **Answer: (D).** Need $a - 2 = 5 \Rightarrow a = 7$. Constants $3 \neq 7$, so the statement is false for every x : no solution. ($a \neq 7$ gives exactly one solution.)
9. **Answer: 13.** $6x + 3a = bx + 21$: $b = 6$, $3a = 21 \Rightarrow a = 7$. $a + b = 13$.
10. **Answer: (B).** $8t = 96 \Rightarrow t = 12$: girls 36, boys 60; difference 24 (equivalently $2t$).
11. **Answer: (D).** $2A = h(b_1 + b_2) \Rightarrow \frac{2A}{h} = b_1 + b_2 \Rightarrow b_1 = \frac{2A}{h} - b_2$.
12. **Answer: (A).** $150 + 45n = 690 \Rightarrow 45n = 540 \Rightarrow n = 12$.
13. **Answer: (B).** $4.5t \leq 55 \Rightarrow t \leq 12.\bar{2}$. Whole trips: 12 (check: $60 - 54 = 6 \geq 5$; 13 trips leave $1.5 < 5$).
14. **Answer: (B).** $-10 < -2x \leq 6 \Rightarrow 5 > x \geq -3$, i.e. $-3 \leq x < 5$: $-3, \dots, 4$, which is 8 integers.
15. **Answer: 24.** $0.4x + 4 = 0.25x + 7.6 \Rightarrow 0.15x = 3.6 \Rightarrow x = 24$.
16. **Answer: 170.** $2(w + (2w - 3)) = 54 \Rightarrow 3w - 3 = 27 \Rightarrow w = 10$. Length 17. Area = $10 \cdot 17 = 170$.
17. **Answer: (C).** $a^2 - 9 = 0 \Rightarrow a = \pm 3$. $a = 3$: $0 = 6$, no solution. $a = -3$: $0 = 0$, infinitely many. Other a : exactly one. So 2 values.
18. **Answer: (C).** $a + b = 3a - 3b \Rightarrow 4b = 2a \Rightarrow \frac{a}{b} = 2$.
19. **Answer: 143.** $4n + 12 = 2(n + 6) + 18 \Rightarrow 2n = 18 \Rightarrow n = 9$: 9, 11, 13, 15 (sum $48 = 30 + 18$). Middle product $11 \cdot 13 = 143$.
20. **Answer: $\frac{4}{3}$.** $4(2x - 1) - 3(x + 2) = 2x - 6 \Rightarrow 8x - 4 - 3x - 6 = 2x - 6 \Rightarrow 5x - 10 = 2x - 6 \Rightarrow x = \frac{4}{3}$.
21. **Answer: (D).** (D): $x \leq -2$ or $x > 1$. (A) covers all reals ($x \geq -2$ or $x < 1$). (B) is empty ($x \leq -2$ and $x > 1$). (C) has open/closed endpoints swapped ($x < -2$ or $x \geq 1$).
22. **Answer: 4.** $x + k = 12 \Rightarrow x = 12 - k$. Set $12 - k = 2k \Rightarrow k = 4$.
23. **Answer: 4.** $(k - 1)x = k - 3 \Rightarrow x = \frac{k-3}{k-1} = 1 - \frac{2}{k-1}$. Integer iff $(k - 1) \mid 2$: $k - 1 \in \{\pm 1, \pm 2\} \Rightarrow k \in \{2, 0, 3, -1\}$ ($x = -1, 3, 0, 2$). Sum = $2 + 0 + 3 - 1 = 4$. ($k = 1$ gives $0 = -2$, no solution, excluded.)
24. **Answer: 71.5.** $(x + 2) = (3x - 2)$: $x = 2$, sides 4, 4, -6 : **invalid**. $(x + 2) = (5x - 16)$: $x = 4.5$, sides 6.5, 11.5, 6.5: valid ($13 > 11.5$), perimeter 24.5. $(3x - 2) = (5x - 16)$: $x = 7$, sides 9, 19, 19: valid, perimeter 47. Sum = 71.5.